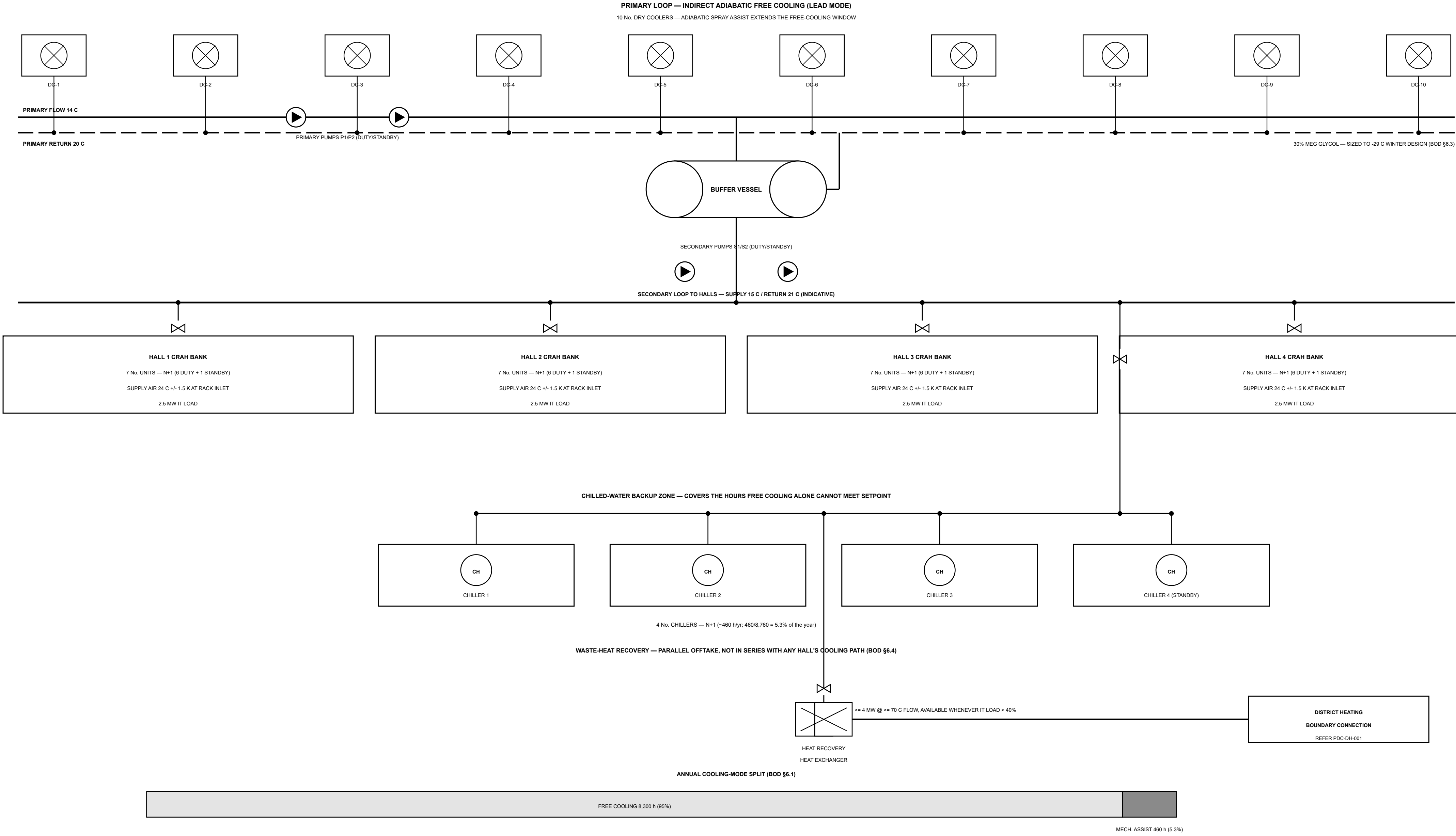




- LEGEND**
- fan Dry cooler / air-cooled unit
 - Pump (duty/standby)
 - Isolation valve
 - Heat exchanger
 - Buffer vessel
 - Junction — pipework connection
 - (no) Crossing — NOT connected

- NOTES**
- Indirect adiabatic free cooling is the primary (lead) mode; outside air meeting the wet-bulb condition is used directly through the indirect coil, with adiabatic spray assist extending the free-cooling window (BOD §6.1).
 - Chilled-water backup, served from the shared backup-chiller zone, covers only the hours neither free cooling nor adiabatic assist alone can meet the 24 C +/- 1.5 K setpoint (~460 h/yr of the design 8,760 h).
 - Heat recovery is a PARALLEL offtake from the backup-chiller zone, not in series with any hall's primary cooling path, so a heat-recovery fault cannot remove cooling capacity from an operating hall (BOD §6.4).
 - Pipework is pre-insulated, welded and pressure-tested; glycol concentration is set to the -29 C Finnish winter design condition to prevent freezing in any external run.
 - Adiabatic make-up water is metered and capped at 12,000 m3/year site-wide — the controlling WUE constraint (BOD §6.2), well inside the 0.25 L/kWh guarantee.
 - Each hall's CRAH bank is N+1 and concurrently maintainable: any single unit is removable without loss of cooling capacity or IT load (ER §4.1).

DESIGN CONDITIONS (BOD §6 / ER §4.1, §4.2)	
DESIGN PARAMETER	VALUE
Summer external design condition	+29 C dry bulb / +19 C wet bulb
Winter external design condition	-29 C
White space supply air (rack inlet)	24 C +/- 1.5 K
Humidity	20-70% RH, non-condensing
Free-cooling hours	>> 8,300 h/yr (~95% of the year)
Water use effectiveness (WUE)	<= 0.25 L/kWh guarantee; design ~0.137 L/kWh
Adiabatic make-up water cap	12,000 m3/yr site-wide
Heat recovery duty	>> 4 MW th @ ~> 70 C, whenever IT load > 40%

NOT TO SCALE

PROJECT
POLARIS DC-1 — AVIAPOLIS DATA CAMPUS, PHASE...

EMPLOYER
POLARIS DIGITAL INFRASTRUCTURE OY

CONTRACTOR
NORDBUILD OY

DRAWING
COOLING SCHEMATIC & HEAT RECOVERY

DISCIPLINE
MECHANICAL

DWG No	REV	DATE	SCALE
PDC-DWG-013	A	24.02.2026	NTS
SYNTHETIC DEMONSTRATION DATA — NOT A REAL PROJECT			